

UNIT 3 – LINKED LIST

TOPIC: CANTEEN ORDER PROCESSING USING LINKED LIST

1. Introduction

A linked list is a linear data structure consisting of nodes connected using pointers. Each node stores data and a link to another node.

This project implements a Canteen Order Processing system using three types of linked lists: Singly Linked List, Doubly Linked List, and Circular Linked List. It is developed using C++ and allows users to insert, display, search, and delete food orders.

2. Problem Statement

Develop a Canteen Order Processing System using a menu-driven C++ program that implements Singly Linked List, Doubly Linked List, and Circular Linked List to manage student food orders. The system should perform insertion, deletion, searching, and traversal operations to efficiently handle canteen orders.

3. Objectives

1. To implement Singly Linked List, Doubly Linked List, and Circular Linked List.
2. To perform insertion and deletion operations.
3. To search for a specific food order using Order ID.
4. To display orders using different traversal methods.
5. To understand dynamic memory allocation using pointers.
6. To apply linked list concepts to a practical application.

4. Data Structures Used

4.1 Singly Linked List (SLL)

A Singly Linked List consists of nodes containing data and a pointer to the next node. The last node points to NULL.

Operations performed:

- Insertion at the beginning.
- Traversal and display.
- Searching for an order.
- Deletion of an order.

4.2 Doubly Linked List (DLL)

A Doubly Linked List consists of nodes containing data, a pointer to the previous node, and a pointer to the next node.

Operations performed:

- Insertion at the beginning.
- Insertion at the end.
- Forward traversal.
- Backward traversal.
- Searching for an order.
- Deletion of an order.

4.3 Circular Linked List (CLL)

A Circular Linked List is a linked list in which the last node points back to the first node instead of NULL.

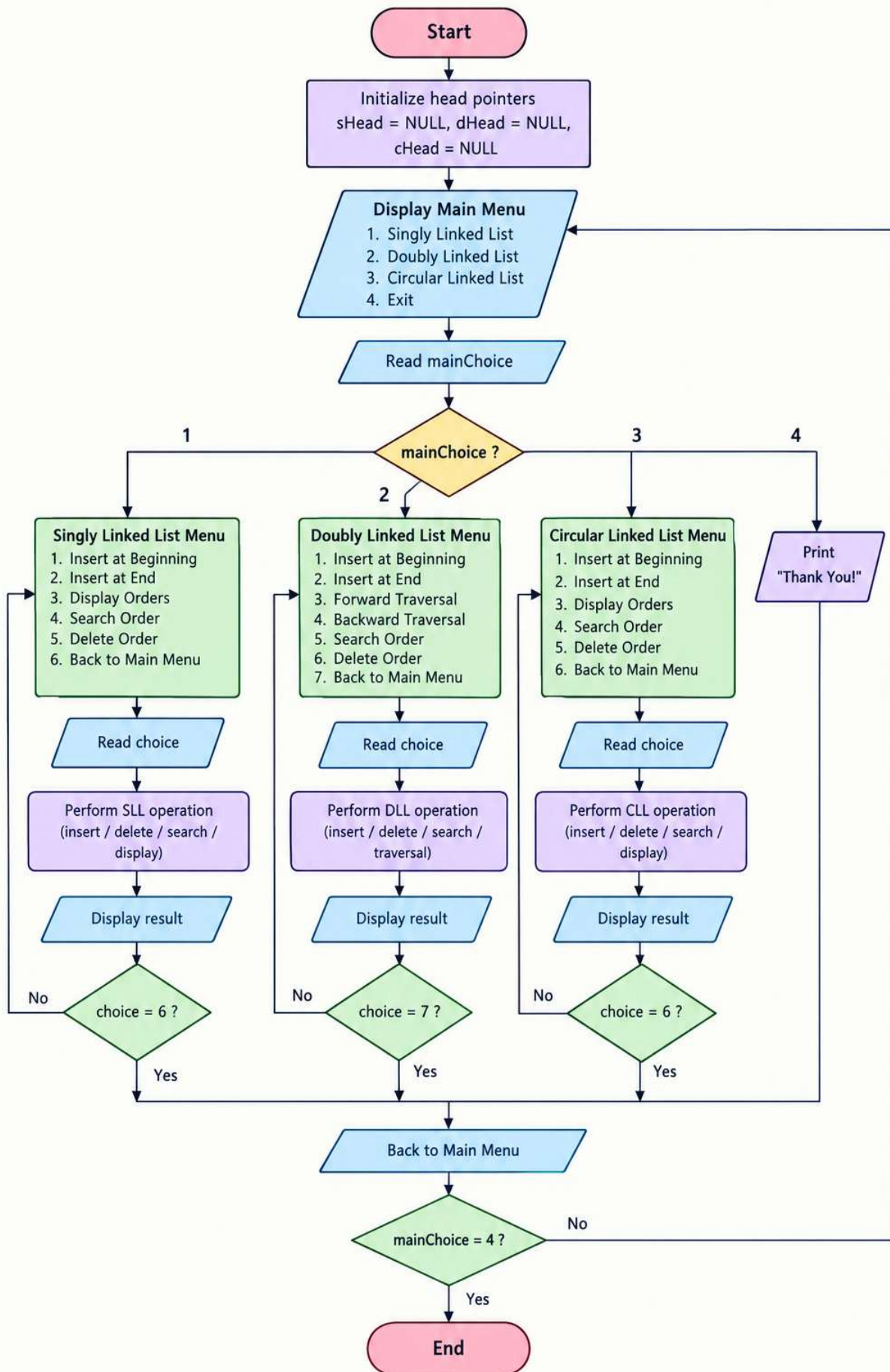
Operations performed:

- Insertion at the beginning.
- Insertion at the end.
- Circular traversal.
- Searching for an order.
- Deletion of an order.

5. Algorithm

1. Start the program.
2. Initialize the head pointers of all three linked lists to NULL.
3. Display the main menu.
4. Select Singly, Doubly, or Circular Linked List.
5. Display the selected linked list's operations.
6. Accept the user's choice.
7. Perform the selected insertion, deletion, searching, or traversal operation.
8. Display the result.
9. Repeat the operations until the user returns to the main menu.
10. Exit the program when the Exit option is selected.
11. Stop.

6. Flowchart



7. Working of the Project

The program uses a menu-driven approach to manage food orders.

Each order contains an Order ID, Student Name, and Food Item. The user selects a linked list and performs the required operation.

The Singly Linked List allows forward traversal. The Doubly Linked List allows forward and backward traversal. The Circular Linked List connects its last node back to the first node.

Dynamic memory allocation is used to create new nodes, and the delete operator is used to free memory when an order is removed.

8. Programming Language and Concepts

Programming Language: C++

Concepts used:

- Structures
- Pointers
- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Dynamic Memory Allocation
- Functions
- Loops
- Conditional Statements
- Switch Statement

9. Applications

1. Managing food orders in a college canteen.
2. Maintaining a list of pending orders.
3. Searching for orders using their identification numbers.
4. Demonstrating different linked list operations.
5. Understanding dynamic data management.

10. Advantages

1. Nodes can be created dynamically as required.
2. Insertion and deletion can be performed without shifting other elements.
3. Different linked list types can be studied in one project.
4. Orders can be searched using their Order IDs.
5. The menu-driven interface makes the program easy to use.

11. Conclusion

Thus, we implemented a Canteen Order Processing system in C++ using Singly Linked List, Doubly Linked List, and Circular Linked List. The project demonstrates insertion, deletion, searching, and traversal operations and helps us understand the practical applications of linked lists.